

Compendium of Formulas, Identities, and Conditions

Chapter 4: Complex Numbers and Quadratic Equations

Authoritative Mathematical Reference Manual

1 Introduction and Core Definitions

1.1 Motivation and Extension of Number System

The real number equation $x^2 + 1 = 0$ has no solution in the real number system $\mathbb{R}$ because $x^2 = -1$ and the square of every real number is non-negative ($x^2 \geq 0$). To enable solutions for equations where the discriminant $D = b^2 - 4ac < 0$, the real number system is extended to the complex number field $\mathbb{C}$.

Definition of Imaginary Unit i

The symbol i (iota) denotes the square root of -1 :

$$i = \sqrt{-1} \implies i^2 = -1$$

The imaginary unit i is a fundamental solution to the equation $x^2 + 1 = 0$.

1.2 Definition and Components of a Complex Number

A number of the form $a + ib$, where $a, b \in \mathbb{R}$, is defined as a **complex number** z .

$$z = a + ib \quad \text{or} \quad z = a + bi \quad (a, b \in \mathbb{R})$$

- **Real Part of z :** $\text{Re}(z) = a$
- **Imaginary Part of z :** $\text{Im}(z) = b$
- **Purely Real Condition:** $\text{Im}(z) = 0 \implies z = a$
- **Purely Imaginary Condition:** $\text{Re}(z) = 0 \wedge \text{Im}(z) \neq 0 \implies z = ib$

1.3 Equality Condition for Complex Numbers

Two complex numbers $z_1 = a + ib$ and $z_2 = c + id$ are equal ($z_1 = z_2$) if and only if their respective real and imaginary parts are identical:

$$z_1 = z_2 \iff \text{Re}(z_1) = \text{Re}(z_2) \wedge \text{Im}(z_1) = \text{Im}(z_2) \iff a = c \text{ and } b = d$$

2 Algebraic Operations and Field Axioms

Let $z_1 = a + ib$ and $z_2 = c + id$ be any two complex numbers.

2.1 Addition of Complex Numbers

The sum $z_1 + z_2$ is defined as:

$$z_1 + z_2 = (a + c) + i(b + d)$$

Properties of Addition:

1. **Closure Law:** $z_1 + z_2 \in \mathbb{C}$ for all $z_1, z_2 \in \mathbb{C}$.
2. **Commutative Law:** $z_1 + z_2 = z_2 + z_1$.
3. **Associative Law:** $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ for all $z_1, z_2, z_3 \in \mathbb{C}$.
4. **Additive Identity:** The zero complex number $0 = 0 + i0$ satisfies $z + 0 = z$.
5. **Additive Inverse:** For $z = a + ib$, the additive inverse is $-z = -a + i(-b)$, satisfying $z + (-z) = 0$.

2.2 Difference of Complex Numbers

The difference $z_1 - z_2$ is defined as the sum of z_1 and the additive inverse of z_2 :

$$z_1 - z_2 = z_1 + (-z_2) = (a - c) + i(b - d)$$

2.3 Multiplication of Complex Numbers

The product $z_1 z_2$ is defined as:

$$z_1 z_2 = (ac - bd) + i(ad + bc)$$

Properties of Multiplication:

1. **Closure Law:** $z_1 z_2 \in \mathbb{C}$ for all $z_1, z_2 \in \mathbb{C}$.
2. **Commutative Law:** $z_1 z_2 = z_2 z_1$.
3. **Associative Law:** $(z_1 z_2) z_3 = z_1 (z_2 z_3)$.
4. **Multiplicative Identity:** The complex number $1 = 1 + i0$ satisfies $z \cdot 1 = z$.
5. **Multiplicative Inverse:** For every non-zero complex number $z = a + ib \neq 0$ (i.e., $a^2 + b^2 \neq 0$), there exists an inverse $z^{-1} = \frac{1}{z}$ given by:

$$z^{-1} = \frac{a}{a^2 + b^2} + i \left(\frac{-b}{a^2 + b^2} \right) = \frac{\bar{z}}{|z|^2} \quad \text{such that} \quad z \cdot z^{-1} = 1$$

6. Distributive Laws:

$$(a) \ z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3 \quad (b) \ (z_1 + z_2)z_3 = z_1 z_3 + z_2 z_3$$

2.4 Division of Complex Numbers

For $z_2 \neq 0$, the quotient $\frac{z_1}{z_2}$ is defined as:

$$\frac{z_1}{z_2} = z_1 \cdot \frac{1}{z_2} = (a + ib) \left[\frac{c}{c^2 + d^2} + i \left(\frac{-d}{c^2 + d^2} \right) \right] = \frac{(ac + bd) + i(bc - ad)}{c^2 + d^2}$$

3 Powers of i and Radical Constraints

3.1 Integral Powers of i

The powers of i exhibit a cyclic period of 4:

Positive Powers	Negative Powers	General Form ($k \in \mathbb{Z}$)	Value
$i^1 = i$	$i^{-1} = \frac{1}{i} = -i$	i^{4k+1}	i
$i^2 = -1$	$i^{-2} = \frac{1}{i^2} = -1$	i^{4k+2}	-1
$i^3 = -i$	$i^{-3} = \frac{1}{i^3} = i$	i^{4k+3}	$-i$
$i^4 = 1$	$i^{-4} = \frac{1}{i^4} = 1$	i^{4k}	1

Sum of Four Consecutive Powers

For any integer $n \in \mathbb{Z}$, the sum of four consecutive integral powers of i is identically zero:

$$i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$$

3.2 Square Roots of Negative Real Numbers

For any positive real number $a > 0$:

$$\sqrt{-a} = \sqrt{a} \cdot \sqrt{-1} = \sqrt{a} i$$

The square roots of $-a$ are $\sqrt{a} i$ and $-\sqrt{a} i$. By standard convention, the symbol $\sqrt{-a}$ represents the principal root $\sqrt{a} i$.

Critical Constraint on Radical Product $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$

The radical identity $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$ holds **true** if:

- $a \geq 0$ and $b \geq 0$
- $a > 0$ and $b < 0$, or $a < 0$ and $b > 0$

The identity is **FALSE** if both $a < 0$ and $b < 0$:

$$\sqrt{a} \cdot \sqrt{b} \neq \sqrt{ab} \quad \text{when } a < 0 \text{ and } b < 0$$

Contradiction Proof: If $\sqrt{a}\sqrt{b} = \sqrt{ab}$ held for $a = b = -1$, then:

$$i^2 = \sqrt{-1}\sqrt{-1} = \sqrt{(-1)(-1)} = \sqrt{1} = 1$$

which contradicts the fundamental definition $i^2 = -1$.

4 Standard Algebraic Identities in $\mathbb{C}$

For all complex numbers $z_1, z_2 \in \mathbb{C}$, the following algebraic expansion formulas hold identically:

1. Square of Sum:

$$(z_1 + z_2)^2 = z_1^2 + 2z_1z_2 + z_2^2$$

2. Square of Difference:

$$(z_1 - z_2)^2 = z_1^2 - 2z_1z_2 + z_2^2$$

3. Cube of Sum:

$$(z_1 + z_2)^3 = z_1^3 + 3z_1^2z_2 + 3z_1z_2^2 + z_2^3$$

4. Cube of Difference:

$$(z_1 - z_2)^3 = z_1^3 - 3z_1^2z_2 + 3z_1z_2^2 - z_2^3$$

5. Difference of Squares:

$$z_1^2 - z_2^2 = (z_1 + z_2)(z_1 - z_2)$$

5 Modulus and Conjugate of a Complex Number

Let $z = a + ib$ be a complex number.

5.1 Definitions

- **Modulus of z :** Non-negative real distance defined by:

$$|z| = \sqrt{a^2 + b^2} \geq 0$$

- **Conjugate of z :** Complex number formed by reversing the imaginary sign:

$$\bar{z} = a - ib$$

5.2 Fundamental Product Relation

The product of a complex number and its conjugate yields the square of its modulus:

$$z \cdot \bar{z} = (a + ib)(a - ib) = a^2 - (ib)^2 = a^2 + b^2 = |z|^2$$

This provides the operational relation for multiplicative inverse:

$$z^{-1} = \frac{\bar{z}}{|z|^2} \quad (z \neq 0)$$

5.3 Theorems and Operational Properties

For any complex numbers $z_1, z_2 \in \mathbb{C}$:

#	Property	Condition / Note
1	$ z_1 z_2 = z_1 z_2 $	Universal
2	$\left \frac{z_1}{z_2} \right = \frac{ z_1 }{ z_2 }$	Provided $ z_2 \neq 0$
3	$\overline{z_1 \pm z_2} = \overline{z_1} \pm \overline{z_2}$	Additive linearity
4	$\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$	Multiplicative linearity
5	$\overline{\left(\frac{z_1}{z_2} \right)} = \frac{\overline{z_1}}{\overline{z_2}}$	Provided $z_2 \neq 0$
6	$ z = \overline{z} = -z = -\overline{z} $	Modulus symmetry
7	$z + \overline{z} = 2 \operatorname{Re}(z)$	Real component extraction
8	$z - \overline{z} = 2i \operatorname{Im}(z)$	Imaginary component extraction

6 Argand Plane Representation

6.1 Geometric Mapping

Each complex number $z = x + iy$ corresponds uniquely to an ordered pair $(x, y) \in \mathbb{R}^2$ represented as point $P(x, y)$ in a two-dimensional Cartesian system called the **Argand Plane** or **Complex Plane**.

- **Real Axis:** The horizontal X -axis ($\operatorname{Im}(z) = 0$).
- **Imaginary Axis:** The vertical Y -axis ($\operatorname{Re}(z) = 0$).
- **Modulus $|z|$:** Represents the Euclidean distance from the origin $O(0, 0)$ to $P(x, y)$:

$$OP = \sqrt{x^2 + y^2} = |z|$$

- **Conjugate $\overline{z}$:** Represents the geometric reflection of $P(x, y)$ across the Real Axis to $P'(x, -y)$.

7 Solutions of Quadratic Equations ($D < 0$)

For a standard quadratic equation with real coefficients:

$$ax^2 + bx + c = 0 \quad (a, b, c \in \mathbb{R}, a \neq 0)$$

When the discriminant $D = b^2 - 4ac < 0$, the equation has no real roots. Its complex solutions are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{-(4ac - b^2)}}{2a} = \frac{-b \pm i\sqrt{4ac - b^2}}{2a}$$

The two roots are complex conjugate pairs.